

Randomized Subspace QAOA: Residual-Controlled Active-Subspace Optimization for Multi-Angle QAOA

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Abstract

Multi-angle QAOA assigns independent cost and mixer angles, yielding $d = p(|E| + |V|)$ parameters. Randomized Subspace QAOA (RSQ) optimizes within a matrix-free approximation to an observable-Jacobian row space and monitors discarded sensitivity with probes. A forward-only variant combines finite-difference Jacobian actions with simultaneous perturbation. We establish a gradient row-space identity, a probe-confidence envelope, and local guarantees conditional on the true residual. On 36 paired exact-statevector MaxCut runs over 14 topologies, RSQ at tolerance 10^{-2} reduces optimized dimension by 39.1% at depth one and 69.6% at depth two. Paired approximation-ratio changes are $+0.48 \pm 0.94$ and -4.77 ± 2.47 percentage points, with descriptive uncertainty over 14 topology clusters per depth. All scheduled checks trigger rebuilding, and access accounting shows setup overhead. The results establish compression with a depth-dependent quality trade-off, not adaptive timing, hardware scaling, or query advantage.

1 Introduction

The Quantum Approximate Optimization Algorithm (QAOA) alternates a cost Hamiltonian and a mixer, then adjusts their angles with a classical optimizer [6, 7, 19]. Multi-angle QAOA (ma-QAOA) replaces the two shared angles in each layer by one phase parameter per cost term and one mixer parameter per qubit [10]. This removes a strong tying constraint, but it changes the classical search dimension from $2p$ to

$$d = p(|E| + |V|). \quad (1)$$

The added coordinates are useful only when they create independent changes in measurable observables. Searching all of them indiscriminately can therefore spend optimization effort in directions to which the circuit is locally insensitive.

Prior QAOA dimension-reduction methods act on different mathematical objects. Symmetry-aware tying uses graph automorphisms to identify redundant ma-QAOA angles and has preserved objective quality on many symmetric instances [16]. ANASTAARS instead constructs adaptive random Johnson–Lindenstrauss search subspaces, fits interpolation models, reuses measurements, and adjusts trust regions to estimated noise [5]. Hamiltonian-guided leverage embedding compresses weighted Ising feature matrices assembled from measurement samples before a reduced trust-region search [13]. Kernel PCA learns a nonlinear, cross-instance manifold of promising QAOA parameters from training graphs [2], while low-rank tensor completion reconstructs variational landscapes from partial observations [9].

RSQ targets a narrower object: the local parameter-space right singular subspace of the vector-valued observable Jacobian $\mathbf{J}(\boldsymbol{\theta})$. It accesses that operator through JVPs and VJP (or JVPs alone),

and monitors the residual sensitivity in the orthogonal complement as the anchor moves. This distinction specifies the operator and diagnostic; it does not assert priority for subspace optimization in QAOA. Classical active subspaces are commonly defined from a gradient outer product averaged over an input distribution [3, 4]. We therefore use *local observable-sensitivity subspace* for the row space of a vector-valued Jacobian at one anchor; it is local, observable-dependent, and not a distribution-averaged global active subspace.

RSQ combines an observable-space formulation with adaptive matrix-free randomized linear algebra. The numerical primitive is fixed-tolerance randomized QB, which approximates a linear operator through products, estimates its residual from random probes, and chooses its output rank adaptively [8, 12, 18]. Our contribution is not a new general-purpose QB factorization or a priority claim for low-dimensional QAOA optimization. It is the specific coupling of a local observable-Jacobian operator, matrix-free sensitivity actions, randomized residual monitoring, and recycled basis updates. A forward-only realization replaces adjoint actions during construction, monitoring, and optimization. The experiments count the setup cost introduced by this coupling instead of treating dimension reduction as a proxy for resource savings.

The central empirical question is not whether a subspace can be constructed, but whether the resulting compression preserves optimization quality after its setup cost is counted. We answer that question on a deliberately small, fully released benchmark. The answer is conditional: at depth one, RSQ closely tracks full ma-QAOA while optimizing substantially fewer coordinates; at depth two, stronger compression incurs a measurable quality loss; and on the studied simulator the operator setup is more expensive than direct optimization. This claim boundary is a result, because it separates structural compression from an unsubstantiated hardware-speedup claim.

Contributions.

1. We formulate the per-edge observable Jacobian as a matrix-free operator and prove that its row space contains the local gradient of any weighted MaxCut objective.
2. We give a residual-gated optimizer with a recycled update that can rotate a saturated basis: old directions are augmented by current residual directions and the combined space is recompressed. The public result exposes every residual check, refresh, retained rank, operator count, final iterate and best observed iterate rather than hiding adaptive decisions behind one score.
3. We give an end-to-end forward-only path: finite-difference JVPs construct and refresh the basis, and two-evaluation SPSA updates the reduced coordinates. A dedicated test fails if reverse-mode differentiation is invoked.
4. We release paired results, operator counters, tests, figures, and the script that regenerates every aggregate in this article.

2 Methods

2.1 Observable-space formulation

Let $G = (V, E)$ be an unweighted graph and let $C_e = (I - Z_i Z_j)/2$ for $e = (i, j)$. A depth- p ma-QAOA state is

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{\ell=1}^p \left[\prod_{i \in V} e^{-i\beta_{\ell i} X_i} \prod_{e \in E} e^{-i\gamma_{\ell e} C_e} \right] |+\rangle^{\otimes |V|}. \quad (2)$$

Table 1: Claim–evidence ledger. Each supported statement is paired with its verification path and the boundary beyond which it is not asserted.

Claim	Evidence	Boundary
The scalar gradient is observable-active	Proposition 1; chain rule	Local and observable-dependent
Discarded first-order change is controlled	Theorem 1; true-residual assumption	No global convergence or ratio guarantee
Depth-one compression has a small measured change	18 paired runs aggregated to 14 topology clusters; Fig. 1	Not an equivalence test; exact simulation only
Depth-two compression loses quality	18 paired runs aggregated to 14 topology clusters; Fig. 2	Tested families and hyperparameters only
The release is not query-advantaged	Instrumented counts; Fig. 4	Counters are not additive shot costs

Define the vector of local observables

$$F(\boldsymbol{\theta}) = (\langle \psi(\boldsymbol{\theta}) | C_e | \psi(\boldsymbol{\theta}) \rangle)_{e \in E} \in \mathbb{R}^m, \quad m = |E|, \quad (3)$$

and its Jacobian

$$\mathbf{J}(\boldsymbol{\theta}) = \frac{\partial F}{\partial \boldsymbol{\theta}} \in \mathbb{R}^{m \times d}. \quad (4)$$

For edge weights $w \in \mathbb{R}^m$, the expected cut is $C(\boldsymbol{\theta}) = w^\top F(\boldsymbol{\theta})$.

Proposition 1 (The local objective gradient is active). *At every differentiable point,*

$$\nabla C(\boldsymbol{\theta}) = \mathbf{J}(\boldsymbol{\theta})^\top w \in \text{range}(\mathbf{J}(\boldsymbol{\theta})^\top). \quad (5)$$

Consequently, the steepest first-order change of any weighted sum of the chosen observables lies in the parameter-space row span of $\mathbf{J}$.

Proof. Writing $C(\boldsymbol{\theta}) = \sum_{e=1}^m w_e F_e(\boldsymbol{\theta})$, its j th partial derivative is $\partial_j C = \sum_e w_e \partial_j F_e$. The j th component of $\mathbf{J}^\top w$ is the same sum. Hence $\nabla C = \mathbf{J}^\top w$, which belongs to the range of $\mathbf{J}^\top$ by construction. $\square$

This statement is local. The row space may rotate as optimization moves, which is why a fixed subspace can fail even if it is exact at its construction point. It also gives an algebraic ceiling

$$\text{rank}(\mathbf{J}) \leq \min(m, d). \quad (6)$$

For the observable choice used here, m does not grow with QAOA depth while $d = p(m+n)$ does. Depth can therefore increase the compressible complement even without additional numerical singular-value decay.

2.2 Randomized Subspace QAOA

2.2.1 Matrix-free sensitivity actions

At an anchor $\boldsymbol{\theta}_0$, RSQ exposes two actions without materializing $\mathbf{J}$:

$$\text{jvp}(v) = \mathbf{J}(\boldsymbol{\theta}_0)v, \quad (7)$$

$$\text{vjp}(u) = \mathbf{J}(\boldsymbol{\theta}_0)^\top u. \quad (8)$$

The implementation evaluates the VJP by reverse-mode automatic differentiation. It evaluates a JVP either by automatic differentiation or by the central finite difference

$$\mathbf{J}v = \frac{F(\boldsymbol{\theta}_0 + hv) - F(\boldsymbol{\theta}_0 - hv)}{2h} + O(h^2), \quad (9)$$

with $h = 10^{-4}$. The released tests compare both actions to a dense Jacobian and verify the adjoint identity.

2.2.2 Adaptive basis and residual diagnostics

RSQ factors $A = \mathbf{J}^\top$ as $A \approx \mathbf{Q}B$, where $\mathbf{Q} \in \mathbb{R}^{d \times r}$ has orthonormal columns and $B = \mathbf{Q}^\top A$. Randomized QB expands $\mathbf{Q}$ in blocks until an estimated relative residual falls below ε or the algebraic rank ceiling is reached; the return value distinguishes these stopping conditions. We expose a Frobenius indicator and a power-iteration spectral indicator, following established randomized matrix-free approximation [8, 14, 18]. The optimization variables are $z \in \mathbb{R}^r$ with

$$\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{Q}z. \quad (10)$$

Every $K = 25$ steps, RSQ evaluates the residual at the current point. If the relative residual exceeds $\varepsilon_{\text{refresh}} = 0.05$, it reanchors. A naive recycled implementation can freeze: if the old basis already has the maximum requested rank, no new column can enter even when the residual is large. Our update instead permits a temporary augmented basis, adds current residual directions, and takes the singular value decomposition of the small coefficient matrix B . Rotating by its leading left singular vectors returns the best requested-rank basis available in the augmented space. This step is both implemented and regression tested.

The implementation records the step and value of every randomized residual check, every triggered refresh, the retained-rank trajectory, and forward, JVP and VJP counts separately. It returns both the final iterate used by the released tables and the best objective value observed along the trajectory. This distinction prevents a user-facing deployment from silently discarding a better intermediate point while preserving the final-iterate convention of the paired experiment.

2.2.3 Forward-only construction, refresh and optimization

The forward-only variant selects an r -dimensional trial space by applying JVPs to a Gaussian parameter sketch and solving the resulting small Rayleigh–Ritz problem. Its refresh diagnostic also uses only JVPs. For independent $v_i \sim \mathcal{N}(0, I_d)$, define

$$\hat{\rho}_F^2 = \frac{\sum_{i=1}^s \|\mathbf{J}(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)v_i\|^2}{\sum_{i=1}^s \|\mathbf{J}v_i\|^2}. \quad (11)$$

Forward differences evaluate both numerator and denominator actions. The identity $\mathbb{E} \|Mv\|^2 = \|M\|_F^2$ implies that the numerator and denominator converge to the corresponding squared Frobenius norms as s grows. We use $s = 12$. Unlike a spectral indicator, Eq. (11) controls average discarded sensitivity; a missed worst direction remains a failure mode.

Reduced optimization also avoids reverse-mode access. For $L(z) = -C(\boldsymbol{\theta}_0 + \mathbf{Q}z)$ and a Rademacher vector $\delta_t \in \{-1, 1\}^r$, it uses

$$\hat{g}_t = \frac{L(z_t + c\delta_t) - L(z_t - c\delta_t)}{2c} \delta_t, \quad c = 10^{-4}. \quad (12)$$

If L has bounded third derivatives, Taylor expansion and $\mathbb{E}[\delta_{ti}\delta_{tj}] = \mathbf{1}\{i = j\}$ give $\mathbb{E}[\hat{g}_t \mid z_t] = \nabla L(z_t) + O(c^2)$ componentwise [17]. To leading order, its i th component is

$$\hat{g}_{ti} = \partial_i L(z_t) + \sum_{j \neq i} \partial_j L(z_t) \delta_{ti} \delta_{tj} + O(c^2), \quad (13)$$

so a single noiseless estimate has $\text{Var}(\hat{g}_{ti} \mid z_t) = \sum_{j \neq i} (\partial_j L(z_t))^2 + O(c^2)$: decreasing c removes leading finite-difference bias but not the cross-coordinate perturbation variance. If the two objective evaluations instead have independent, zero-mean errors of variance σ^2 , their contribution to each component variance is $\sigma^2/(2c^2)$. Shot noise therefore creates the usual bias–variance trade-off; correlated sampling, repeated perturbations, and a scheduled c may help, but require a separate analysis. The released exact-statevector experiment has $\sigma^2 = 0$. Equation (12) costs two objective evaluations independently of r , but it is not an exact gradient, and the finite run supplies no asymptotic SPSA convergence claim.

Proposition 2 (Moment identity for the forward residual diagnostic). *Let $P = \mathbf{Q}\mathbf{Q}^\top$, let $R = \mathbf{J}(\mathbf{I} - P)$, and let $v_1, \dots, v_s$ be independent standard Gaussian vectors. For*

$$\hat{a}_s = \frac{1}{s} \sum_{i=1}^s \|Rv_i\|^2, \quad (14)$$

we have

$$\mathbb{E}[\hat{a}_s] = \|R\|_F^2, \quad \text{Var}(\hat{a}_s) = \frac{2}{s} \|R^\top R\|_F^2. \quad (15)$$

The analogous identities hold for the denominator of Eq. (11) with R replaced by $\mathbf{J}$.

Proof. Set $H = R^\top R$. For a standard real Gaussian vector v , $\mathbb{E}[vv^\top] = \mathbf{I}$, so $\mathbb{E}[v^\top H v] = \text{tr}(H) = \|R\|_F^2$. Isserlis’ identity gives $\mathbb{E}[(v^\top H v)^2] = \text{tr}(H)^2 + 2 \text{tr}(H^2)$; therefore its variance is $2 \text{tr}(H^2) = 2 \|R^\top R\|_F^2$. The s summands are independent, so the variance of their mean is divided by s . Replacing R by $\mathbf{J}$ proves the denominator statement. $\square$

Proposition 2 justifies the diagnostic as a consistent randomized estimator. The following result states what can be concluded at a finite probe count without assuming a stable rank.

Proposition 3 (Finite-probe confidence envelope). *Let $a = \|R\|_F^2 > 0$, $b = \|\mathbf{J}\|_F^2 > 0$, and let $\hat{\rho}_F = (\hat{a}_s/\hat{b}_s)^{1/2}$ be formed from Gaussian quadratic-form averages as in Eq. (11). For $\delta \in (0, 1)$ set $t = 2/\sqrt{s\delta}$. If $t < 1$, then, with probability at least $1 - \delta$,*

$$\sqrt{\frac{1-t}{1+t}} \hat{\rho}_F \leq \frac{\|R\|_F}{\|\mathbf{J}\|_F} \leq \sqrt{\frac{1+t}{1-t}} \hat{\rho}_F. \quad (16)$$

Proof. For either quadratic-form average, Proposition 2 and $\|M^\top M\|_F^2 \leq \|M\|_F^4$ give relative variance at most $2/s$. Chebyshev’s inequality with radius t fails with probability at most $2/(st^2) = \delta/2$ for each of numerator and denominator. A union bound makes both relative bounds simultaneous. Taking the worst ratio and then a square root yields Eq. (16). $\square$

The envelope is intentionally conservative. At the released $s = 12$ and 95% confidence, $t > 1$, so it is non-informative: the observed ratio is a refresh diagnostic, not a run-wise confidence certificate. More than 80 probes are required before this elementary 95% envelope is finite, and a usefully tight interval requires substantially more. The ratio is not unbiased, no finite probe count gives a deterministic upper bound, and central finite differences add the $O(h^2)$ error of Eq. (9). Accordingly, Theorem 1 is conditional on the true operator residual.

Proposition 4 (True-relative-residual bridge). *Let $P = \mathbf{Q}\mathbf{Q}^\top$ be an orthogonal projector and suppose $\|\mathbf{J}(\boldsymbol{\theta})\|_F > 0$. Define the true relative Frobenius residual*

$$\rho_F(\boldsymbol{\theta}, P) = \frac{\|\mathbf{J}(\boldsymbol{\theta})(\mathbf{I} - P)\|_F}{\|\mathbf{J}(\boldsymbol{\theta})\|_F}. \quad (17)$$

For $C = w^\top F$ and $g = \nabla C(\boldsymbol{\theta})$,

$$\|g - Pg\| \leq \rho_F(\boldsymbol{\theta}, P) \|\mathbf{J}(\boldsymbol{\theta})\|_F \|w\|. \quad (18)$$

Proof. Proposition 1, symmetry of P , and $P^2 = P$ give $(\mathbf{I} - P)g = [\mathbf{J}(\mathbf{I} - P)]^\top w$. Hence

$$\begin{aligned} \|g - Pg\| &\leq \|\mathbf{J}(\mathbf{I} - P)\|_2 \|w\| \\ &\leq \|\mathbf{J}(\mathbf{I} - P)\|_F \|w\| = \rho_F \|\mathbf{J}\|_F \|w\|. \end{aligned}$$

□

Proposition 4 is the bridge from the true relative quantity approximated by Eq. (11) to an objective-level statement. Substituting the finite-probe ratio for ρ_F is not justified without an accompanying probabilistic envelope.

2.2.4 Local consequences of a true residual bound

Theorem 1 (True-residual projected-step discrepancy). *Let $P = \mathbf{Q}\mathbf{Q}^\top$ and suppose $\|\mathbf{J}(\boldsymbol{\theta})(\mathbf{I} - P)\|_2 \leq \varepsilon$. Let $C = w^\top F$ be twice continuously differentiable on a convex neighborhood containing the segments from $\boldsymbol{\theta}$ to both $\boldsymbol{\theta} + \Delta$ and $\boldsymbol{\theta} + P\Delta$, and suppose $\|\nabla^2 C(\xi)\|_2 \leq L_C$ throughout that neighborhood. Then*

$$|C(\boldsymbol{\theta} + \Delta) - C(\boldsymbol{\theta} + P\Delta)| \leq \|w\| \varepsilon \|\Delta\| + L_C \|\Delta\|^2. \quad (19)$$

Proof. Taylor's theorem gives $C(\boldsymbol{\theta} + \Delta) = C(\boldsymbol{\theta}) + \nabla C(\boldsymbol{\theta})^\top \Delta + r_1$ with $|r_1| \leq L_C \|\Delta\|^2/2$, and the analogous expansion along $P\Delta$ has $|r_2| \leq L_C \|P\Delta\|^2/2 \leq L_C \|\Delta\|^2/2$. Proposition 1 makes the linear difference $w^\top \mathbf{J}(\boldsymbol{\theta})(\mathbf{I} - P)\Delta$. Cauchy-Schwarz and the assumed operator-norm residual bound give magnitude at most $\|w\| \varepsilon \|\Delta\|$. Adding $|r_1| + |r_2|$ proves the claim. □

The theorem does not give global convergence or an approximation-ratio guarantee. It states exactly what a true residual bound controls: the first-order change discarded at the current anchor.

Theorem 2 (One exact projected-gradient ascent step). *Let $P = \mathbf{Q}\mathbf{Q}^\top$, let $g = \nabla C(\boldsymbol{\theta})$, and suppose ∇C is L -Lipschitz on the segment from $\boldsymbol{\theta}$ to $\boldsymbol{\theta}^+ = \boldsymbol{\theta} + \alpha Pg$, with $L > 0$ and $0 < \alpha \leq 1/L$. Then*

$$C(\boldsymbol{\theta}^+) \geq C(\boldsymbol{\theta}) + \alpha \left(1 - \frac{L\alpha}{2}\right) \|Pg\|^2 \geq C(\boldsymbol{\theta}) + \frac{\alpha}{2} \|Pg\|^2. \quad (20)$$

If $\|\mathbf{J}(\boldsymbol{\theta})\|_F > 0$, the omitted gradient component additionally satisfies Eq. (18).

Proof. The smoothness inequality applied to $s = \boldsymbol{\theta}^+ - \boldsymbol{\theta} = \alpha Pg$ gives

$$\begin{aligned} C(\boldsymbol{\theta} + s) &\geq C(\boldsymbol{\theta}) + g^\top s - \frac{L}{2} \|s\|^2 \\ &= C(\boldsymbol{\theta}) + \alpha \|Pg\|^2 - \frac{L\alpha^2}{2} \|Pg\|^2, \end{aligned}$$

where $g^\top Pg = \|Pg\|^2$ because P is an orthogonal projector. The first inequality in Eq. (20) follows; the step-size condition gives the second. The final statement is Proposition 4. □

Theorem 2 is deliberately a one-step result for exact projected-gradient ascent with a fixed projector. The released two-sided path uses Adam, the forward-only path feeds the estimator in Eq. (12) to Adam, and both paths may replace P after a randomized residual check. The theorem is therefore not a convergence guarantee for Adam, SPSA, finite-probe residual estimates, or a moving randomized basis.

Algorithm. The implemented loop is:

1. construct $\mathbf{Q}$ at θ_0 to tolerance ε or the rank ceiling, and record the stopping status;
2. update z for the objective $-C(\theta_0 + \mathbf{Q}z)$, using Adam for the two-sided path or a two-evaluation Rademacher SPSA estimate for the forward-only path [11, 17];
3. every K steps, estimate the current residual;
4. when it exceeds threshold, reanchor, augment and recompress the recycled basis, reset z , and continue.

Standalone observable evaluations, JVPs and VJP are instrumented as overlapping categories; ranks, residual checks and refreshes are recorded separately.

2.3 Access and computational complexity

Let r be the retained rank, b the QB block size, q the number of expansion blocks, s_0 the initial norm-probe count and s the residual-probe count. Ignoring pruning, the Frobenius two-sided build uses $s_0 + (q+1)s + qb$ VJPs and qb JVPs. A two-sided refresh of an existing rank- r basis uses r JVPs to form its coefficient matrix and $2s$ VJPs for the residual ratio. This describes the diagnostic itself; a triggered rebuild incurs its own norm probes, residual probes, coefficient updates and any expansion blocks. The forward-only build uses $k = \min(d, r + \text{oversampling})$ JVPs; a standalone forward-only residual check uses $2s$ JVPs. A central-difference JVP costs two observable-map evaluations. Each forward-only SPSA optimization step uses two objective evaluations, followed by one final-iterate evaluation at the end of the run. These exact categories are emitted in the released CSV; they overlap and must not be added as if they were disjoint physical-shot costs.

For the bundled exact simulator, one observable-map evaluation takes $O(p(m+n)2^n)$ arithmetic and stores $O((m+1)2^n)$ numbers because all per-edge cut indicators are precomputed. This backend is a small-instance reference, not a large-scale simulator. On hardware, one computational-basis sample can update all commuting MaxCut edge observables, but shot complexity depends on the desired joint precision and noise model; neither a shot bound nor a circuit-depth advantage is claimed here. The matrix-free callback interface permits external backends without changing the QB core.

2.4 Experimental protocol

We test unweighted MaxCut on three graph families: random 3-regular graphs, Erdos–Renyi graphs with edge probability 0.5, and rings. The grid contains $n \in \{8, 10\}$, $p \in \{1, 2\}$, and three restart identifiers: 36 paired optimizer runs. Random regular and Erdos–Renyi generation yields 12 distinct graphs; the two rings repeat across restart identifiers, giving 14 unique topologies and 28 topology–depth clusters. Graph generation, angle initialization and randomized linear algebra use separately recorded seeds (0–2, 100000–100002, and 200000–200002, respectively). A canonical edge list and SHA-256-derived graph identifier accompany every row. The simulator is a pure-PyTorch, complex-128 statevector implementation [15]; an optional PennyLane backend supplies an independent cross-check [1]. Brute-force MaxCut gives the exact denominator for the approximation ratio.

Every optimizer receives 100 steps at learning rate 0.05. Full ma-QAOA, the fixed-rank oracle and both randomized variants share the same run-specific initial angle vector. The symmetry baseline begins at its orbit-wise mean projection, so it is derived from the same vector while satisfying its tying constraints. Adam is used except for the forward-only SPSA update [11, 17]. We compare:

- full ma-QAOA;
- angles tied by graph-automorphism orbits;
- a rank-8 dense-SVD oracle fixed at the initial Jacobian (its materialization is recorded separately rather than disguised as a matrix-free action);
- two-sided adaptive RSQ at $\varepsilon \in \{10^{-1}, 10^{-2}\}$;
- forward-only rank-8 RSQ.

The primary outcome is expected-cut approximation ratio. Parameter count is the dimension optimized after construction. Paired differences subtract the full ma-QAOA ratio on the same graph, initialization and sketch identifier. Repeated restarts of an identical topology are averaged within each topology–depth cluster before analysis. We report the mean and sample standard deviation across 14 cluster means per depth for raw ratios, and the mean $\pm$ two standard errors across the corresponding paired cluster means. These are descriptive summaries, not independent-instance hypothesis tests; no absence of significance is interpreted as equivalence.

3 Results

3.1 Depth one: compression with a small paired change

At $p = 1$, full ma-QAOA optimizes 23.7 parameters on average and attains 0.862 ± 0.053 . RSQ at $\varepsilon = 10^{-2}$ optimizes 14.7 parameters, a 39.1% reduction, and attains 0.867 ± 0.051 . The paired change is $+0.48 \pm 0.94$ percentage points. The two-standard-error bar spans zero, but the experiment was not designed as a formal equivalence trial; this grid does not resolve a directional average change at depth one.

The rank-8 fixed-subspace oracle reaches 0.764 ± 0.051 , a paired change of -9.77 ± 2.09 points. Forward-only rank-8 RSQ reaches 0.798 ± 0.055 , improving on the fixed oracle but remaining 6.39 ± 1.98 points below full ma-QAOA. The comparison is consistent with a benefit from a moving subspace, but it does not isolate refresh from differences in construction and optimization.

3.2 Depth two: the failure boundary becomes visible

At $p = 2$, full ma-QAOA optimizes 47.4 parameters on average and attains 0.945 ± 0.044 . RSQ optimizes the same 14.7-dimensional observable-controlled subspace as at depth one, now a 69.6% reduction. At tolerance 10^{-2} its ratio is 0.897 ± 0.054 , a paired change of -4.77 ± 2.47 points. The added layer grows the ambient parameter space without growing the observable count, but the more aggressive compression no longer preserves the full optimizer’s quality.

The forward-only rank-8 variant loses 14.73 ± 2.65 points at depth two. The fixed rank-8 oracle loses 18.07 ± 2.54 points. These results rule out a blanket claim that a very small initial sensitivity space is sufficient for ma-QAOA. They do not identify which component of the moving-subspace pipeline causes the difference and may indicate a need for richer observables at higher depth.

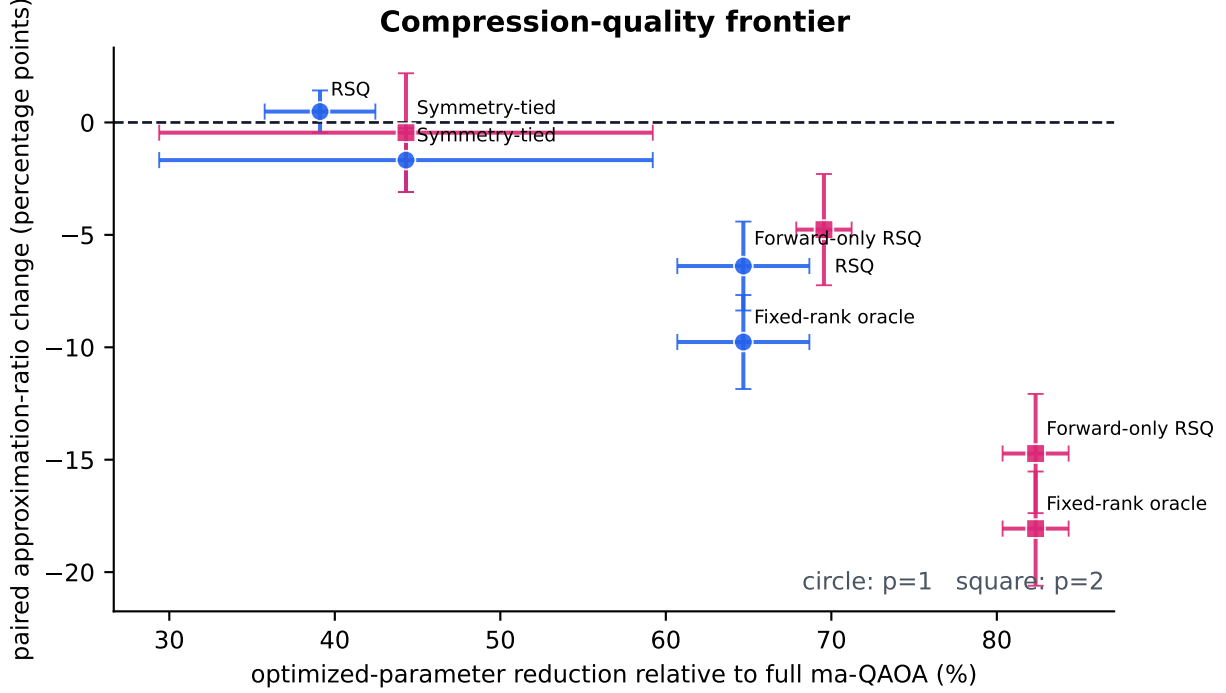


Figure 1: Compression-quality frontier. Horizontal coordinates are optimized parameter reduction relative to full ma-QAOA. Vertical coordinates are paired approximation-ratio changes; error bars show two standard errors. The main RSQ point at $p = 1$ approaches the zero-change line, whereas stronger compression and depth two expose a larger quality cost.

3.3 The residual indicator identifies algebraic, not extra numerical, rank

Across the tested tolerances, two-sided RSQ retains 14.7 directions on average, the average number of edge observables. Tightening the tolerance from 0.1 to 0.01 does not change the mean retained rank, although the paired quality summaries differ modestly. Five of 144 builds at tolerance 0.01 and three of 144 builds at tolerance 0.1 reached the algebraic rank ceiling before their finite-probe indicator met the requested tolerance. Thus this benchmark does not show additional numerical low-rank decay beyond the algebraic ceiling in Eq. (6). What grows with depth is the complement between m and d , not a newly discovered decay of the nonzero spectrum.

3.4 Setup cost prevents a query-efficiency claim

At the primary tolerance 0.01, the two-sided method averages 749 forward- F evaluations, 134 JVPs, and 481 VJP at $p = 1$, compared with 101 forward evaluations and 100 VJP for the full baseline under the simulator’s accounting. At $p = 2$ the corresponding RSQ counts increase to 867, 155, and 556. The forward-only path uses no VJP, as intended, but requires about 488–489 forward evaluations and 143–144 JVPs. These categories are instrumentation counters, not additive physical-shot counts; a JVP by central difference contains forward evaluations already recorded by the lower-level counter.

4 Discussion

The experiments support a narrower and more useful conclusion than the original resource-efficiency hypothesis. At depth one, the observable row space removes roughly two fifths of the optimized coordi-

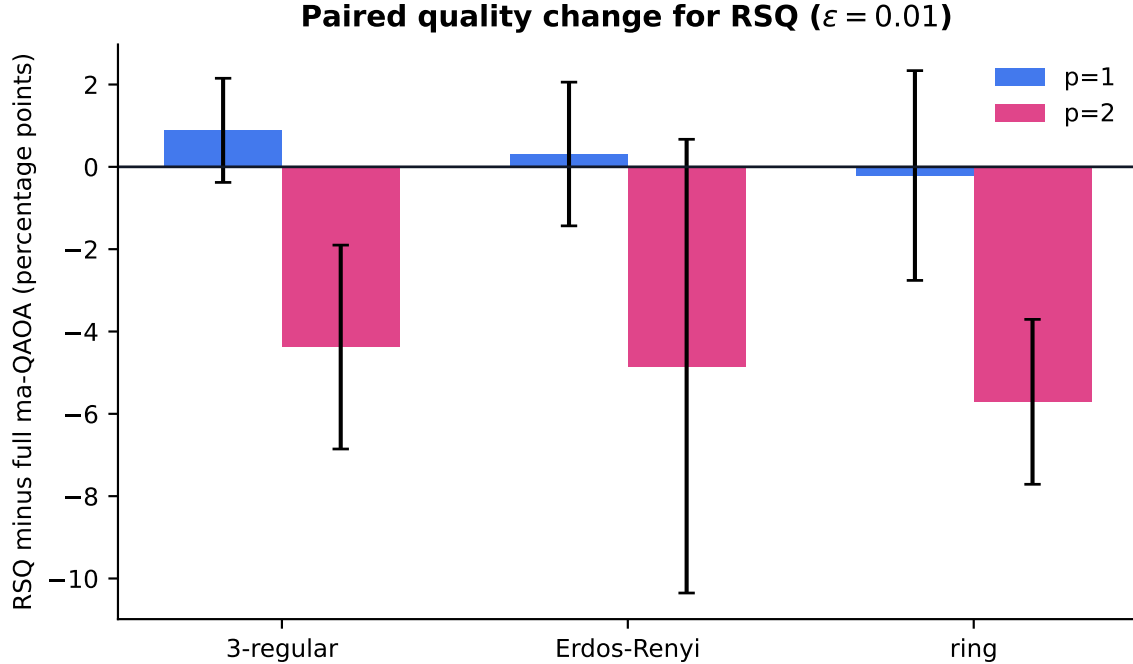


Figure 2: Paired quality change for RSQ at $\varepsilon = 0.01$, separated by graph family. Bars show means and error bars show two standard errors. The depth-two loss is present across all three tested families; wide intervals on some families reflect the small released grid.

nates with little resolved change in mean quality. At depth two, the same observable dimension removes roughly seven tenths of the coordinates but loses about five approximation-ratio points. Parameter compression is therefore real, while quality preservation is depth-dependent.

The fixed-rank oracle underperforms the adaptive pipeline, which is consistent with a local row space that changes along the trajectory. That comparison does not, however, isolate the refresh gate: basis construction, rank, and optimizer also differ. Moreover, 324 of 324 scheduled residual checks across the two tolerances and the forward-only variant exceeded the threshold and triggered a rebuild. The experiment therefore evaluates a periodically refreshed implementation; it does not show that the gate saved rebuilds or selected better refresh times. A matched-rank comparison among no-refresh, fixed-schedule, and threshold-gated variants is needed to identify that effect. The regression suite separately checks that recycling a saturated basis can rotate rather than freezing its old span.

The forward-only path establishes feasibility but not parity. It avoids every VJP, including during refresh, and outperforms the equally sized fixed oracle. Its remaining quality loss can arise from restricting the Rayleigh–Ritz search to a random trial space, finite-difference error, single-perturbation SPSA variance, and the use of an average Frobenius drift diagnostic. Because the released objectives are exact statevector expectations, they do not test the $1/c^2$ measurement-noise amplification in the SPSA estimator. Hardware studies should compare larger sketches, common-random-number shot estimators, repeated perturbations, gain schedules, and worst-direction forward indicators.

For the fixed scalar objective $C = w^\top F$ used here, the weight vector w is known and a single VJP returns $\mathbf{J}^\top w = \nabla C$ directly. Constructing a basis for the larger range of $\mathbf{J}^\top$ therefore has no demonstrated single-objective gradient-query benefit. Such a basis could be amortized across multiple weight vectors, objectives or related instances, but the released benchmark tests none of those use cases. Likewise, the two-evaluation SPSA cost in Eq. (12) is independent of reduced rank, but ordinary full-space SPSA also uses two objective evaluations per step. A full-space SPSA baseline is absent, so the

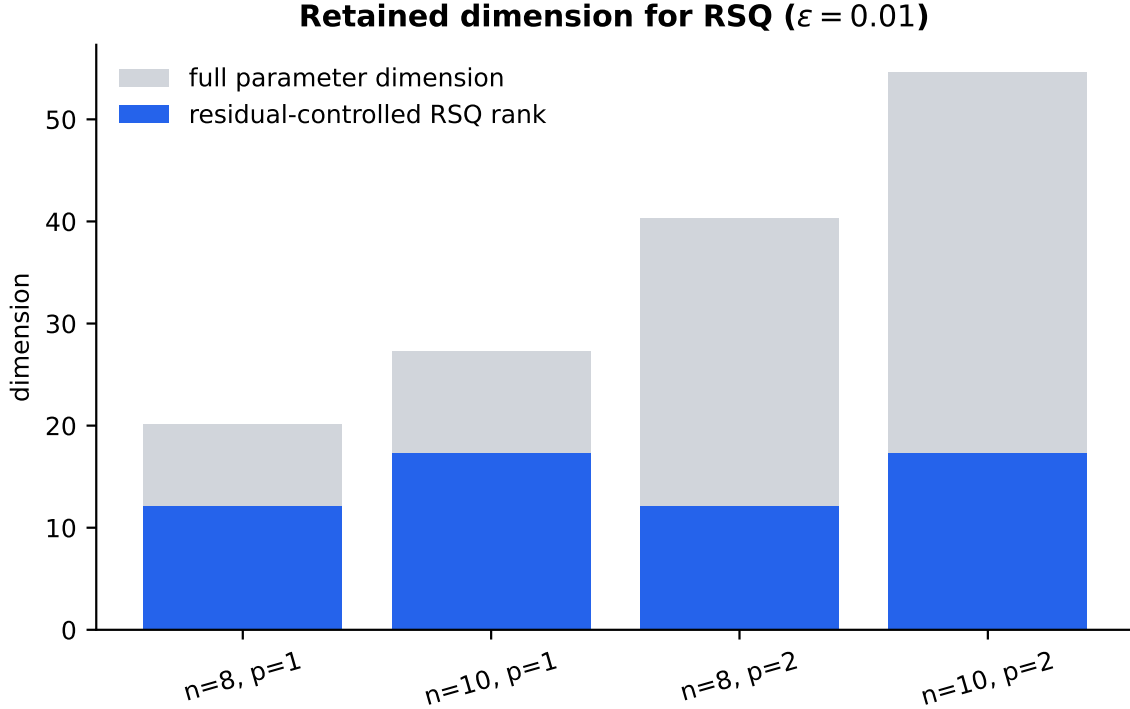


Figure 3: Ambient ma-QAOA dimension and residual-controlled RSQ rank, averaged across graph families and seeds. The observable count controls the retained dimension while the ambient parameter count doubles with depth.

forward-only results establish parameter-space compression and zero-VJP execution, not an objective-query advantage.

Most importantly, fewer optimized coordinates do not automatically mean fewer quantum queries. The released small-instance counts go in the opposite direction because basis construction is expensive. A genuine query advantage would require amortizing a basis across related instances, using hardware-native batching, or reaching a scale at which full-gradient cost grows faster than the matrix-free sketch. Those are testable future conditions, not conclusions of the present data.

5 Limitations

The benchmark contains only 36 paired runs on 14 unique topologies, $n \leq 10$, depths one and two, and exact statevector objectives. It does not model shot noise, device noise, compilation, queue latency, or weighted spin glasses. The graph families are standard but narrow. One hundred optimizer steps may favor some parameterizations over others, and no hyperparameter sweep is used to optimize each baseline separately. For the fixed-rank oracle, dense-Jacobian materialization is included in wall time and recorded as a separate event, but no hardware-query equivalent is assigned; it is therefore primarily a quality reference. The randomized residual ratios are finite-probe estimators and can underestimate the true residual; no deterministic or run-wise coverage guarantee is claimed for the chosen probe count. All 324 of 324 released refresh checks triggered, so no empirical saving from residual gating is established. The projected-ascent theorem does not cover the Adam or SPSA updates used in the experiment, and the exact-statevector setting does not test shot-noise bias-variance trade-offs. The study also omits multi-weight amortization and a full-space SPSA control, both required to test the two potential query-saving mechanisms described above. Finally, the observable vector contains per-edge

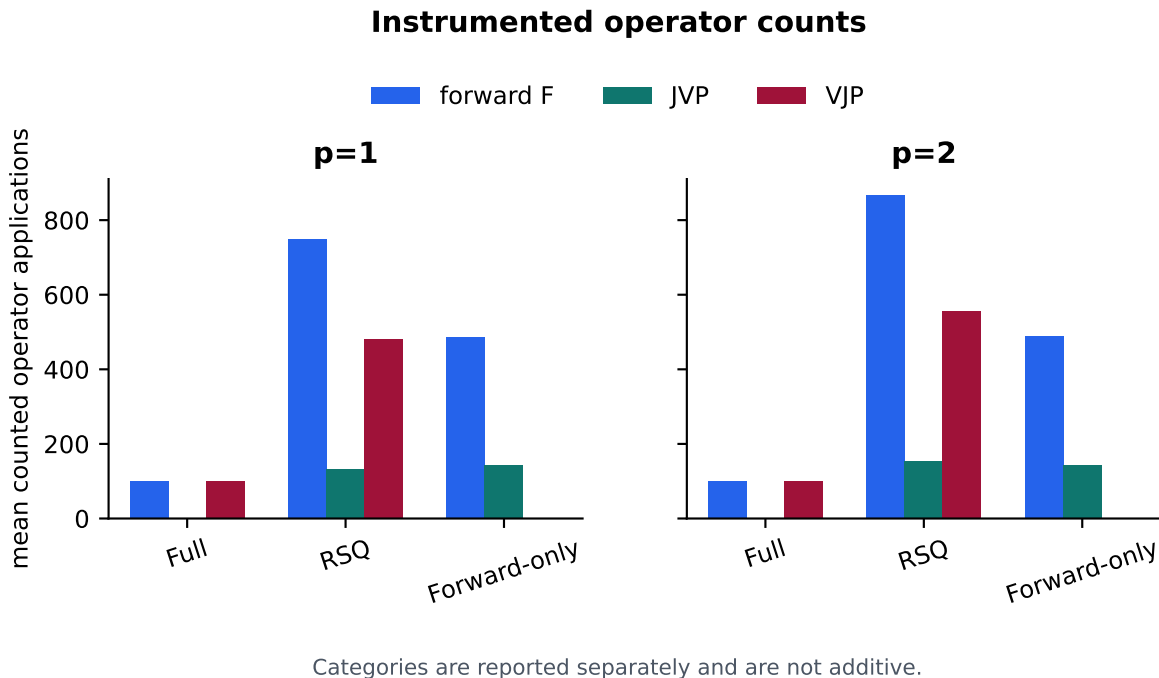


Figure 4: Mean instrumented operator counts. Categories are shown separately and are not additive. On these small exact simulators, subspace discovery and refresh cost more operator applications than direct full-space optimization.

costs; adding mixer observables, commutators, or layer-resolved features could raise rank but improve high-depth fidelity.

6 Reproducibility

The canonical package is the repository’s `code/` directory. A clean environment can install and verify it with

```
python -m pip install -e 'code[dev]'
python -m pytest code/tests -q
cd code
python experiments/run_experiment.py \
    --config experiments/configs/maxcut_small.yaml
python experiments/summarize_results.py
python experiments/validate_release.py
```

The experiment runner writes 216 CSV rows and records separate graph, initialization and sketch seeds; canonical edge lists and graph hashes; complete objective trajectories and final parameters; every residual value, refresh step and basis-build status; deterministic-thread settings; environment versions; and SHA-256 fingerprints of the implementation, experiment runner and effective configuration. The summarizer reads only that CSV and regenerates the four figures, Table 2, and a machine-readable JSON summary. Tests cover the statevector against a dense reference, JVP/VJP consistency, fixed-tolerance factorization, spectral residual estimation, basis recycling, the forward-only zero-VJP invariant, best-iterate and residual-audit reporting, invalid-control rejection, baselines, the command-line JSON con-

Table 2: Released benchmark: 36 paired runs on 14 unique topologies and 28 topology–depth clusters. Approximation ratios are mean $\pm$ sample standard deviation across cluster means. Paired changes are relative to full ma-QAOA on the same run and are mean $\pm$ two standard errors across paired cluster means.

Method	Depth	Parameters	Approx. ratio	Paired change
Full ma-QAOA	1	23.7	0.862 ± 0.053	–
Symmetry-tied	1	14.4	0.845 ± 0.065	-1.67 ± 1.42 pp
Fixed-rank oracle	1	8.0	0.764 ± 0.051	-9.77 ± 2.09 pp
RSQ ($\varepsilon = 0.01$)	1	14.7	0.867 ± 0.051	$+0.48 \pm 0.94$ pp
RSQ ($\varepsilon = 0.1$)	1	14.7	0.862 ± 0.051	-0.03 ± 0.79 pp
Forward-only RSQ	1	8.0	0.798 ± 0.055	-6.39 ± 1.98 pp
Full ma-QAOA	2	47.4	0.945 ± 0.044	–
Symmetry-tied	2	28.7	0.940 ± 0.064	-0.45 ± 2.64 pp
Fixed-rank oracle	2	8.0	0.764 ± 0.048	-18.07 ± 2.54 pp
RSQ ($\varepsilon = 0.01$)	2	14.7	0.897 ± 0.054	-4.77 ± 2.47 pp
RSQ ($\varepsilon = 0.1$)	2	14.7	0.906 ± 0.058	-3.88 ± 1.84 pp
Forward-only RSQ	2	8.0	0.797 ± 0.053	-14.73 ± 2.65 pp

tract, and the optional PennyLane cross-check. No test-count claim is hard-coded because collection varies with optional backends.

7 Data and code availability

The row-level CSV, machine-readable summary, generated tables and figure source data are included with the accompanying repository. The JSON summary records the CSV SHA-256 digest, protocol, sampling units, environment, refresh audit, and hashes of all generated display artifacts. The canonical Python source, tests and exact regeneration commands are released under the MIT License in the same repository. No external or restricted dataset is used. This working release does not yet have an archival DOI; a versioned repository release should be deposited before formal submission.

8 Conclusion

RSQ turns the local observable Jacobian of ma-QAOA into an adaptive optimization space without forming the Jacobian. Its residual has a precise role: the true operator residual bounds discarded first-order sensitivity, while its randomized estimate decides when the space is updated. On the released benchmark, this mechanism preserves depth-one quality under substantial parameter compression, reveals a depth-two compression–quality tradeoff, and carries nontrivial setup cost. These findings establish a fact-checkable baseline for future randomized-subspace quantum optimization and define the evidence required before stronger query-efficiency claims are made.

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